

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459885.77008 +/- 0.00085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1399 +/- 0.0047
Transit depth (Rp/Rs)^2	1.96 +/- 0.13 [%]
Orbital Inclination (inc)	88.29 +/- 0.94
Airmass coefficient 1 (a1)	1.1434 +/- 0.0054
Airmass coefficient 2 (a2)	-0.119 +/- 0.02
Scatter in the residuals of the lightcurve fit is	0.47 %
Best Comparison Star	None
Optimal Aperture	4.49
Optimal Annulus	11.49
Transit Duration (day)	0.1284 +/- 0.0013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1399 ± 0.0047 vs 0.1489 (deviation 1.51σ)
Duration fit vs published	0.1284 d vs 0.1304 d (deviation 1.22σ)
Mid-time O–C	3.4 min
Depth SNR	23.5
Residual scatter	0.47 %

Light curve

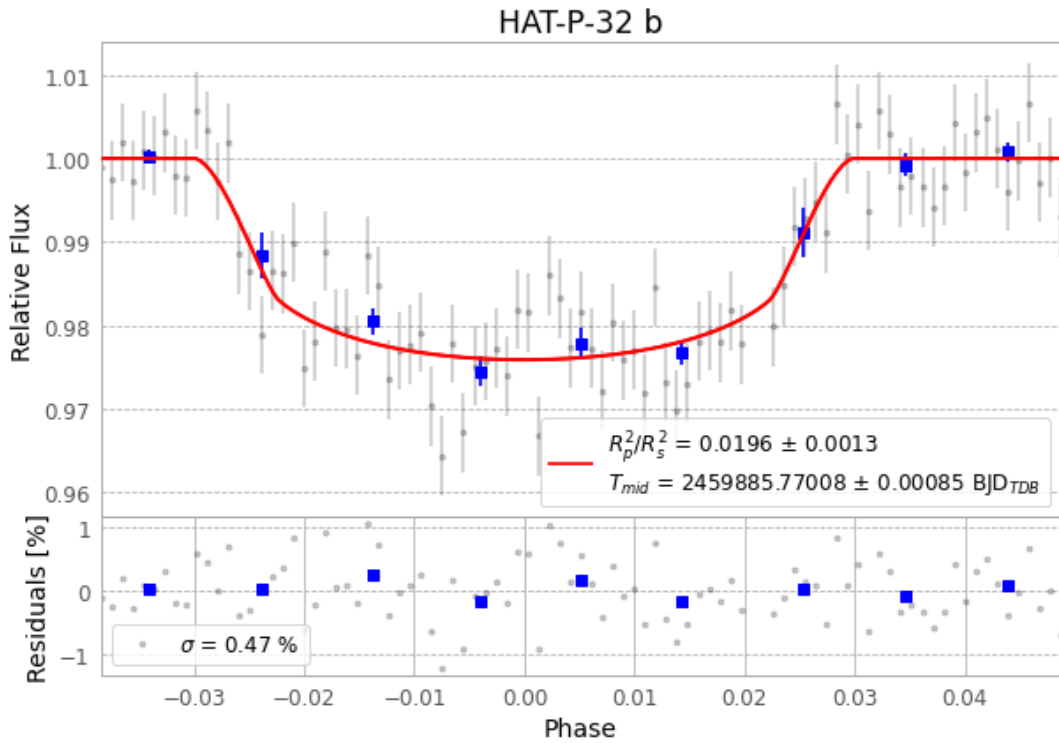


Figure 1 — FinalLightCurve_HAT-P-32 b_2022-11-01.png (SHA-256 8832221f0abcf65b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [496, 130], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 56ce144a6d787c47...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

91 FITS frames (60 MB), HATP-32221102042115.FITS → HATP-32221102085115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-221102.log (b1767f37d894...), FinalLightCurve_HAT-P-32 b_2022-11-01.png (8832221f0abc...), FinalLightCurve_HAT-P-32 b_2022-11-01.csv (5ce7e0eeaef1...)

Compute resources

Wall time 195.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:57Z.